8 Scan Design Handoff

Learning Objectives

After completing this lab, you should be able to:

- Name the 3 types of DFTC output files downstream tools require specifically related to scan testing.
- Write out the final test protocol and a scandef file.

Lab Duration:
30 minutes

Overview

Mapped Flow:

1. Read the gate-level design and test protocol into DFTC.
2. Specify scan constraints and preview the scan architecture that will result from applying them.
3. Insert the scan chain.
4. Hand off the design and related files for use by downstream tools; exit.

Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question, check the back portion of this lab for help.

File Locations

All files for this lab except for designs and libraries are located in the directory **`lab8_export`**.

Directory Structure

`lab8_export`	Current working directory
`analyzed`	Intermediate acs_read_hdl files
`logs`	Session log files
`unmapped`	Unmapped protocol
`mapped`	Gate-level netlist
`mapped_scan`	Gate-level scan design netlist
`reports`	DFTC reports
`tmax`	Files for downstream tools
`scripts`	Constraint and run scripts
`../rtl/ORCA_init/vhdl`	Design files
`../libs`	Library files

Lab 8

Instructions

Given a mapped design, export from DFTC the files required by downstream tools such as ATPG and P&R.

Task 1. Execute the Mapped Flow and Export Files

1. Change directories to the project directory **lab8_export**.

```
UNIX% cd lab8_export
```

2. Invoke DFTC.

```
dc_shell
```

3. Read the gate-level design and test protocol into DFTC.

```
s scripts/4read_gate_and_protocol.tcl
```

4. Specify scan constraints and preview the scan architecture that will result ~~from applying them by using the script provided.~~

即
5preview_dft.tcl

```
s scripts/settings_insert_dft.tcl
preview_dft
```

insert_dft_configuration
定义信号和path

5. Insert the scan chain.

```
s scripts/6insert_dft.tcl
```

```
   Uncollapsed Stuck Fault Summary Report
-----------------------------------------------
fault class                     code   #faults
------------------------------  ----  ---------
Detected                         DT       68835
Possibly detected                PT          70
Undetectable                     UD        1417
ATPG untestable                  AU        5768
Not detected                     ND         108
-----------------------------------------------
total faults                              76198
test coverage                            92.10%
-----------------------------------------------
```

Question 1. What is the estimated test coverage for the scan design?

92.10%

6. Complete the script that writes out the design and related files to be used by downstream tools (scripts/7handoff.tcl).

Add to this script:

- The command to save the needed ATPG SPF file (to the `./tmax` directory) that is not currently being saved. write_test_protocol -o ...
- The command to write out the scan chains to a SCANDEF file (to the `mapped_scan` directory). write_scan_def -output ...
- The command to check the SCANDEF check_scan_def

```
# Write a Verilog netlist
set test_stil_netlist_format verilog
write -f verilog -h -o tmax/ORCA_scan.v

#Complete the script that writes out the design and related files to be used by downstream tools.
#Add to this script:
write_test_protocol -o ./tmax/STIL.spf ; #Writes a STIL test protocol file.
write_scan_def -output ./mapped_scan/SCANDEF.scandef ; #Generates SCANDEF scan chain information for performing scan chain reordering in the physical implementation flow.
check_scan_def ; #Performs scan chain structural consistency checking based either on the scan chain information stored as an attribute in the current .ddc file, or an ASCII SCANDEF file.

# Write the ddc design
write -format ddc -hierarchy -output mapped_scan/ORCA.ddc
```

```
MacroDefs {
    "test_setup" {
        W "_default_WFT_";
        C {
            "all_inputs" = \r11 N;
            "all_outputs" = \r17 X;
            "all_bidirectionals" = \r44 Z;
        }
        V {
            "conf_ena" = 0;
            "pclk" = 0;
            "prst_n" = 1;
            "sdr_clk" = 0;
            "sys_clk" = 0;
        }
        V {
            "conf" = 1;
            "conf_ena" = 1;
            "pclk" = P;
        }
        V {
            "conf" = 0;
        }
        V {
            "conf" = 1;
        }
        V {
            "conf_ena" = 0;
            "pclk" = 0;
        }
        V {
        }
    }
```

```
unix%  vi scripts/7handoff.tcl
```

Execute the handoff script and examine that additional file.

```
s scripts/7handoff.tcl
```

Question 2. Is the custom initialization sequence still there? What has changed about it?

....

....

The test protocol for `preview_dft` contains your scan chain in and out signal declarations.

```
ScanStructures {
    ScanChain "chain0" {
        ScanLength 488;
        ScanIn "pad[0]";
        ScanOut "sd_A[0]";
        ScanEnable "scan_en";
        ScanMasterClock "sdr_clk";
    }
    ScanChain "chain1" {
        ScanLength 488;
        ScanIn "pad[1]";
        ScanOut "sd_A[1]";
        ScanEnable "scan_en";
        ScanMasterClock "pclk" "sdr_clk" "sys_clk";
    }
    ScanChain "chain2" {
        ScanLength 488;
        ScanIn "pad[2]";
        ScanOut "sd_A[2]";
        ScanEnable "scan_en";
        ScanMasterClock "pclk";
    }
    ScanChain "chain3" {
        ScanLength 488;
        ScanIn "pad[3]";
        ScanOut "sd_A[3]";
        ScanEnable "scan_en";
        ScanMasterClock "pclk" "sdr_clk";
    }
    ScanChain "chain4" {
        ScanLength 487;
        ScanIn "pad[4]";
        ScanOut "sd_A[4]";
        ScanEnable "scan_en";
        ScanMasterClock "sdr_clk" "sys_clk";
        ScanInversion 1;
    }
    ScanChain "chain5" {
        ScanLength 487;
        ScanIn "pad[5]";
        ScanOut "sd_A[5]";
        ScanEnable "scan_en";
        ScanMasterClock "sys_clk";
        ScanInversion 1;
    }
}
```

Question 3. What additional information has DFTC added to the ScanStructures section now that the protocol is for a scan inserted design?

....

Question 4. How many scanchains does the SCANDEF file show, and how many partitions were created (use grep or consult the `check_scan_def` report)?

显示详细6条信息。

....

Exit dc_shell

```
exit
```

```
Report : Scan DEF check
Design : ORCA
Version: H-2013.03-SP5
Date   : Wed Sep 12 17:02:02 2018

****************************************

 Information from SCANDEF file:
  Number of SCANCHAINS: 9

 Checking between SCANDEF file and design:
  Total SCANCHAINS checked: 9
  VALIDATED :  9
  FAILED    :  0

Chain name Status  #cells PARTITION          Scan IN        Scan OUT
---------- ------  ------ ---------          -------        --------
1          V       487    sdr_clk_55_55      pad_iopad_0/CIN I_ORCA_TOP/I_SDRAM_IF/mega_
shift_1_reg_29__7_/SD
2_SG1      V       24     sdr_clk_55_55      pad_iopad_1/CIN I_ORCA_TOP/I_SDRAM_IF/LOCKU
P/D
2_SG2      V       462    pclk_45_45         I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/SO
I_ORCA_TOP/I_PCI_CORE/mega_shift_reg_26__6_/SD
3          V       487    pclk_45_45         pad_iopad_2/CIN I_ORCA_TOP/I_PCI_CORE/mega_
shift_reg_56__14_/SD
4_SG1      V       123    pclk_45_45         pad_iopad_3/CIN I_ORCA_TOP/I_RESET_BLOCK/LO
CKUP/D
4_SG2      V       364    sdr_clk_45_45      I_ORCA_TOP/I_RESET_BLOCK/LOCKUP/Q I_ORCA_TO
P/I_SDRAM_IF/mega_shift_0_reg_20__11_/SD
5_SG1      V       255    sdr_clk_45_45      pad_iopad_4/CIN I_ORCA_TOP/I_SDRAM_WRITE_FI
FO/LOCKUP/D
5_SG2      V       231    sys_clk_45_45      I_ORCA_TOP/I_SDRAM_WRITE_FIFO/LOCKUP/Q I_OR
CA_TOP/I_BLENDER/s4_op1_reg_4_/SD
6          V       486    sys_clk_45_45      pad_iopad_5/CIN I_ORCA_TOP/I_SDRAM_WRITE_FI
FO/this_addr_g_int_reg_6_1/SD
```

①TetraMAX用来干什么

Task 2. Verify Exported Files in TetraMAX

1. Change directories to the TetraMAX directory **tmax**.

```
UNIX% cd tmax
```

```
./tmax/STIL.spf
./scripts/orca_mapped.spf
```

2. Inspect the TetraMAX run script. Ensure that the paths for the scan inserted netlist and STIL protocol file referenced in the script match the file written out by DFT Compiler.

```
unix%  vi orca_tmax.tcl
```

```
# Set logfile
set_command noabort
set_messages -log tmax.log -replace -level expert

# Read the library and design
read_netlist libs.v.gz
read_netlist rams.v
read_netlist ORCA_scan.v

# Run build
set_rule b5 warning
run_build

# Run DRC (ensure the name of the SPF matchs what was written from DFTC)
run_drc ORCA_scan.spf

# Add fault adn run ATPG
add_faults -all
run_atpg -auto

# Increase capture cycles to enable fast sequential ATPG
set_atpg -capture_cycles 4
run_atpg -auto
```

3. Run TetraMAX.

```
tmax orca_tmax.tcl
```

Question 5. What is the Test Coverage reported by TetraMAX?

96.72%

Question 6. How does the Test Coverage compare to the estimated coverage reported in DFTC? Can you explain the reason for the difference?

DFTC里面跑没用STIL.spf和test simulation library（?）

```
        Uncollapsed Stuck Fault Summary Report
 -----------------------------------------------
 fault class                     code   #faults
 ------------------------------  ----  ---------
 Detected                         DT      71639
 Possibly detected                PT        757
 Undetectable                     UD       1406
 ATPG untestable                  AU        769
 Not detected                     ND       1297
 -----------------------------------------------
 total faults                             75868
 test coverage                           96.72%
 -----------------------------------------------
            Pattern Summary Report
 -----------------------------------------------
 #internal patterns                         336
     #basic_scan patterns                   243
     #fast_sequential patterns               93
 -----------------------------------------------
            CPU Usage Summary Report
 -----------------------------------------------
 Total CPU time                            9.82
 -----------------------------------------------
```

GUI
Total CPU time ：11.05

ns! You

```
            Pattern Summary Report
 -----------------------------------------------
 #internal patterns                           0
 -----------------------------------------------

        Uncollapsed Stuck Fault Summary Report
 -----------------------------------------------
 fault class                     code   #faults
 ------------------------------  ----  ---------
 Detected                         DT      68835
 Possibly detected                PT         70
 Undetectable                     UD       1417
 ATPG untestable                  AU       5768
 Not detected                     ND        108
 -----------------------------------------------
 total faults                             76198
 test coverage                           92.10%
 -----------------------------------------------
 Information: The test coverage above may be inferior
              than the real test coverage with customized
              protocol and test simulation library.
```